Ant Colony

7) University Examination Timetable Scheduling.

Application : Academic Scheduling.

problem Statement : A university needs to schedule examinations for multiple subjects into available time slots and examination halls implemented ACO to assign examination slots while avoiding student conflicts respecting room capacities and satisfying other scheduling constraints

objective : minimize timetable conflicts and scheduling penalties.

pseudocode :

BEGIN

1. initialize parameters
   Set number of ants (m)
   Set pheromone importance (alpha)
   Set heuristic importance (beta)
   Set evaporation rate (rho)
   Set initial pheromone value (tau)
   Set maximum iterations

2. Define
   List of examinations

take available time slots

Read examination halls

Read Student examination conflicts

Read all capacities

3 Initialize pheromone matrix

Initialize pheromone values for all possible exam-slot hall assignments

4 from iteration = 1 to maxIter do

for each ant k = 1 to m do

select an examination.

identify the suitable time slots and hall based on transition probabilities

Calculate transition probabilities using pheromone and heuristic values

Select time slot and hall based on transition probability

Assign examination to selected time slot and hall

end while

Calculate timetable fitness

Store the best fitness value

5 update pheromones
evaporate existing pheromones
for each ant
Deposit pheromones
6 check Stoping condition
7. Display best examination time table
Display Scheduling penalty.
end.

30/10/26.

→ ~~iteration~~ ~~Best~~ ~~current~~ ~~Best~~
iteration 10 Best fitness : 300.9
iteration 20 Best fitness : 300.9
iteration 30 Best fitness : 300.9
iteration 40 Best fitness : 300.9
iteration 50 Best fitness : 300.9

Best examination timetable
maths time : 9 AM Hall : Hall B
Physics time : 4 PM Hall : Hall C
Chemistry time : 1 PM Hall : Hall C
CS time : 1 PM Hall : Hall A
OS time : 4 PM Hall : Hall B

Best fitness : 300.9 total conflicts : 0